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RESEARCH PAPER

RootBot: root-inspired soft-growing robot for high-curvature directional excavation

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Abstract

Bio-inspired robotics garners a significant interest with an aim to improve performance and efficiency in a variety of geotechnical engineering practices. Particularly, the urban environment calls for a new robotic system that can excavate and navigate freely in the subsurface. This study presents a new soft-growing robotic system, RootBot, which exploits inspiration from plant root growth to excavate with steering and directional excavation capabilities through complicated routes and confined spaces. RootBot is designed by incorporating a locomotion mechanism modeled after the plant root growth mechanisms with excavation and discharge systems inspired by tunnel boring machines. A prototype of the RootBot demonstrates several unique features and abilities including directional excavation with a planar motion, a retractive motion, and a sharp steering ability of more than 90° rotation with a curvature radius less than twice the width of the excavation head. A series of meter-scale experiments demonstrate that the prototype RootBot can effectively excavate and advance in wet-compacted sand. It can steer at high curvature and even pull back and quarter-turn at any point along a straight path. The RootBot is expected to be useful in various geotechnical and urban engineering practices, including directional ground excavation, navigation through a complicated, curved route to pass by unexpected underground obstacles and soil removal in inaccessible and confined spaces for buried pipe repairs.

Keywords Bio-inspired robot · Directional excavation · RootBot · Soft-growing robot

1 Introduction

Biomimetics is a field that actively exploits inspiration from nature to develop new solutions, and it has flourished in various engineering fields, such as aeronautics, medical science, and material science [1, 20]. Pioneering efforts are being made to incorporate biomimetics and bioinspired principles in geotechnical engineering with an aim to improve performance and efficiency in a variety of geotechnical engineering activities such as ground excavation, ground stabilization, and underground infrastructure construction and maintenance [21, 29]. Exemplary attempts include the ground penetration or excavation techniques inspired by moles [19], clams [13, 28], and earthworms [9, 14], and the surface designs of geotechnical structures for selective friction control, inspired by snake skins [22].

Rapid urbanization has accompanied the construction, installation, and operation of underground the infrastructures in subsurface. Various trenchless technologies has been developed from horizontal auger boring, and pneumatic pipe ramming, to large tunnel boring machine (TBM) as underground spaces is often suggested as a preferable option for effective space utilization [6]. However, construction of underground structures still encounters various technical challenges, which include, but are not limited, to road closure for construction and damages to pre-existing structures due to tunneling. Furthermore, the

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maintenance of underground structures, such as buried pipes, buried power transmission cables, sewage tunnel and utility tunnel, also poses challenges due to the limited accessibility. Accordingly, there is a growing demand to develop small and compact tunneling machines or robotic ground excavation systems [3, 16, 19].

Whereas, in recent years, urban environment with various buried structures and sometimes unidentified obstacles calls for directional tunneling or directional excavation. For instance, an excavation robot with a steering ability is required to excavate and advance along a complicated path while detouring an unexpected obstacle. To cope with this demand, there is a wide range of ongoing efforts to develop robotic excavation systems inspired by biological systems, such as clams [13, 28], earthworms [9, 14] and moles [19]. Although those newly proposed robotic systems as well as commercially available excavation techniques have shown their own strength, they also have limitations in path selection and directional steering with high curvature [7, 26].

Meanwhile, the concept of soft-growing robots, inspired by the behavior of plants such as vines, was first proposed by Daisuke et al. [23]. It was later revisited by Hawkes et al. [11] and has since gained significant attention in the field of soft robotics. Researchers have continued to develop and demonstrate these types of robots [4, 10, 15, 17]. This robot has a base, from which it can continuously grow as it expands by adding material at its tip. The unique properties of soft-growing robots such as flexibility and compliance enable them to successfully complete tasks that would be difficult or impossible for traditional, rigid robots to accomplish. The soft-growing robots can adapt to different environments and navigate a space, which makes them well-suited for excavation tasks that require movement through challenging or confined spaces [8].

This study presents a root-inspired soft growing excavation robot, so called RootBot, which has a steering ability and directional excavation capability. This RootBot is composed of three main modules: the locomotive (or propulsion) module inspired by plant root growth, the excavation module with a rotating cutterhead, following the TBM design, and the muck-discharge module with vacuum suction and water circulation. The proposed RootBot is designed to grow freely in space (move forward, retract, steer from a fixed base while excavating). The RootBot is specifically designed to enable directional excavation at high curvatures within a limited radius, by using its ability to grow and steer its body. Furthermore, the growth is hardly affected by side friction as it grows at the tip. This is one of the biggest advantages of the RootBot while the conventional methods such as auger boring or pneumatic pipe ramming methods need to overcome side friction. Finally, a prototype of the RootBot was fabricated and integrated with peripheral equipment. Thereafter, a series of excavation scenarios were tested with the prototype in a meter-scale soil box. The results of the tests on the prototype are analyzed for performance evaluation, and the potential and limitations of the RootBot are discussed.

2 Design and prototyping of RootBot

2.1 Root-inspired design of RootBot

Roots of plants have capability to circumvent obstacles and extend for nutrients. The growth of plant roots is explained with the combination of the three mechanisms: cell differentiation, cell elongation, and cell division, as shown in Fig. 1a [5]. At the tip, the root extends by cell division, i.e., the zone of cell division. At the zone of cell elongation, cells elongate to provide a driving force (or thrust force) to the root cap which enables penetration into soils. The zone of cell differentiation provides a fixed base to transmit the driving force to the root cap. The combination of these three mechanisms allows a root to navigate freely in soils for uptakes of water and nutrients. Herein, the root cap plays two important roles: the obstacle avoidance through steering and the ground penetration through rotation, as shown in Fig. 1b [12].

The proposed soft growing excavation robot, RootBot, is inspired by the growth of plant roots. Inspiration is particularly on the locomotion (or propulsion) of this excavation robot. RootBot extends its body by inflating a pair of soft plastic tube, as shown in Fig. 2. Initially, a soft plastic tube is fed through a roller at a controlled rate (Fig. 2a). At the same time, air pressure inflates the supplied soft plastic tube, which provides the driving force and the ensuing propulsion (Fig. 2b). The tube is stored as uninflated before the roller, but it becomes inflated when fed after the roller. This material feed mechanism allows continuous and almost infinite growth of the robot as long as the tube supply is sustained. The roller fixture can be connected to the excavation module at the frontmost, as shown in Fig. 3. Herein, the robot adopts a pair of soft plastic tubes to which each air pressure can be separately controlled. The locomotion of the excavation module is controlled by the air pressure supply to these two tubes, and thereby, the excavation module can advance, retract, and steer at the frontmost.

Furthermore, the driving force for propulsion (propulsion force or thrust force) of RootBot is applied to the frontmost part, which also mimics the root growth mechanism. Loss of air pressure from the base to the roller is insignificant while the air is supplied from the base.

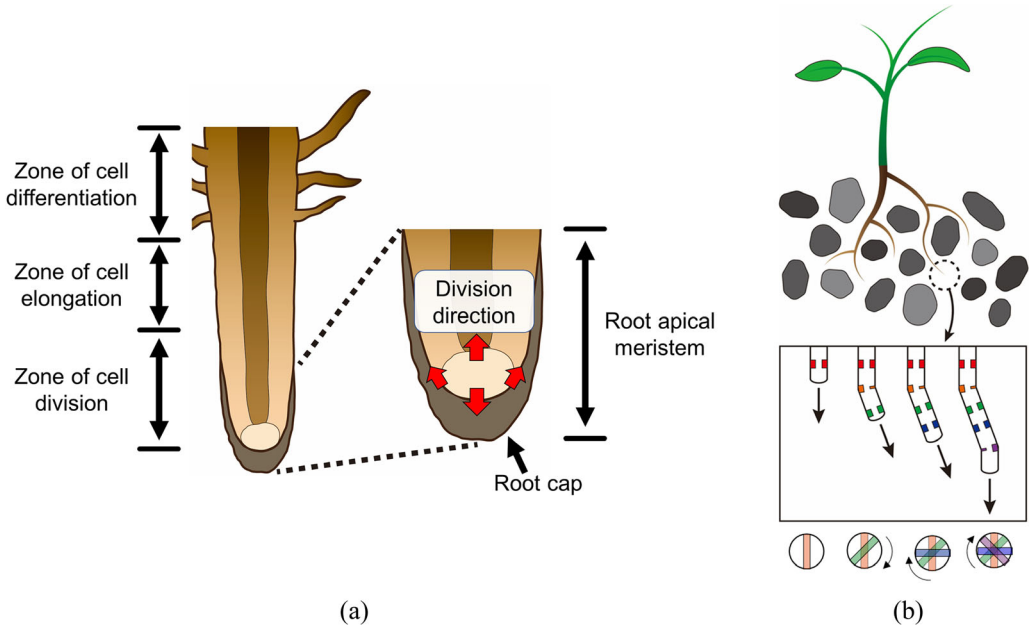


Fig. 1 Bio-inspiration for a soft growing excavation robot: **a** the root growth by cell division (adopted from Bitonti and Chiappetta [5]), and **b** the root growth in soils (modified from Hu et al. [12])

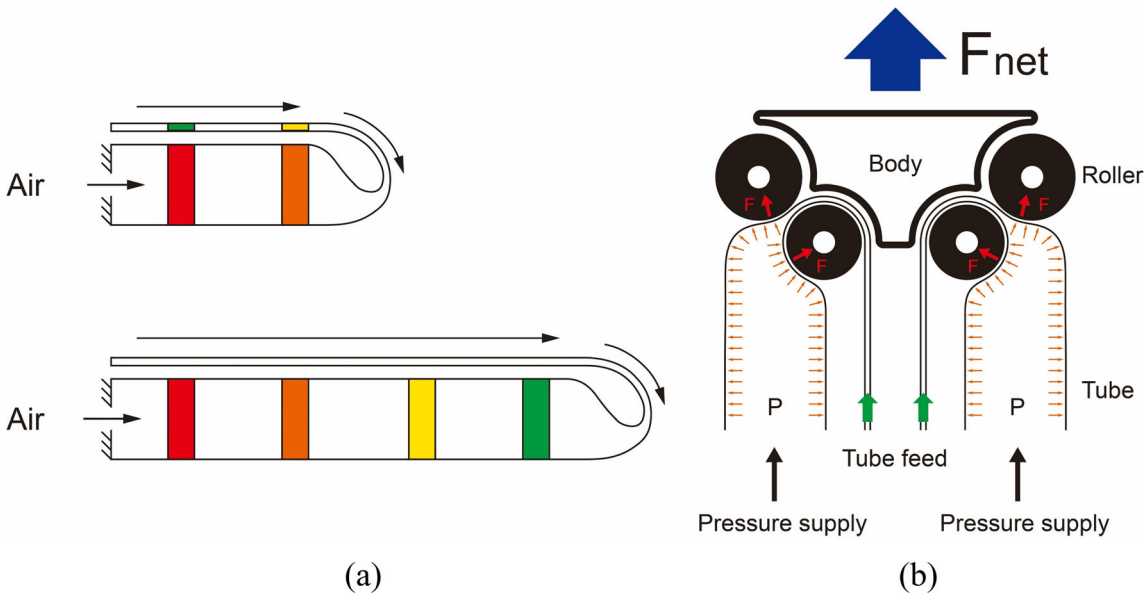


Fig. 2 Mechanism for locomotion of RootBot: **a** tube feed and **b** propulsion by air pressure

Meanwhile, in conventional drilling and ramming-types of excavation, the friction between a drilling auger or pipe and surrounding soils increases as the excavation distance increases. Eventually, this elevated frictional energy loss limits the excavation length. By contrast, the propulsion force in the proposed RootBot can still be efficiently transmitted through tubes by air pressure, even if the inflated tubes are curved, buckled, or in contact with surrounding ground. Furthermore, the frictional loss of propulsion force is expected to be ignorable in RootBot, comparing to the other types of excavation systems due to its unique locomotion, which is a move by growth, and it is not limited by excavation distance and complicated curvy routes.

2.2 Proto-typing of RootBot

This study presents a prototype of RootBot, which consists of three modules: propulsion, excavation, and discharge modules, as shown in Fig. 3. The excavation module in the

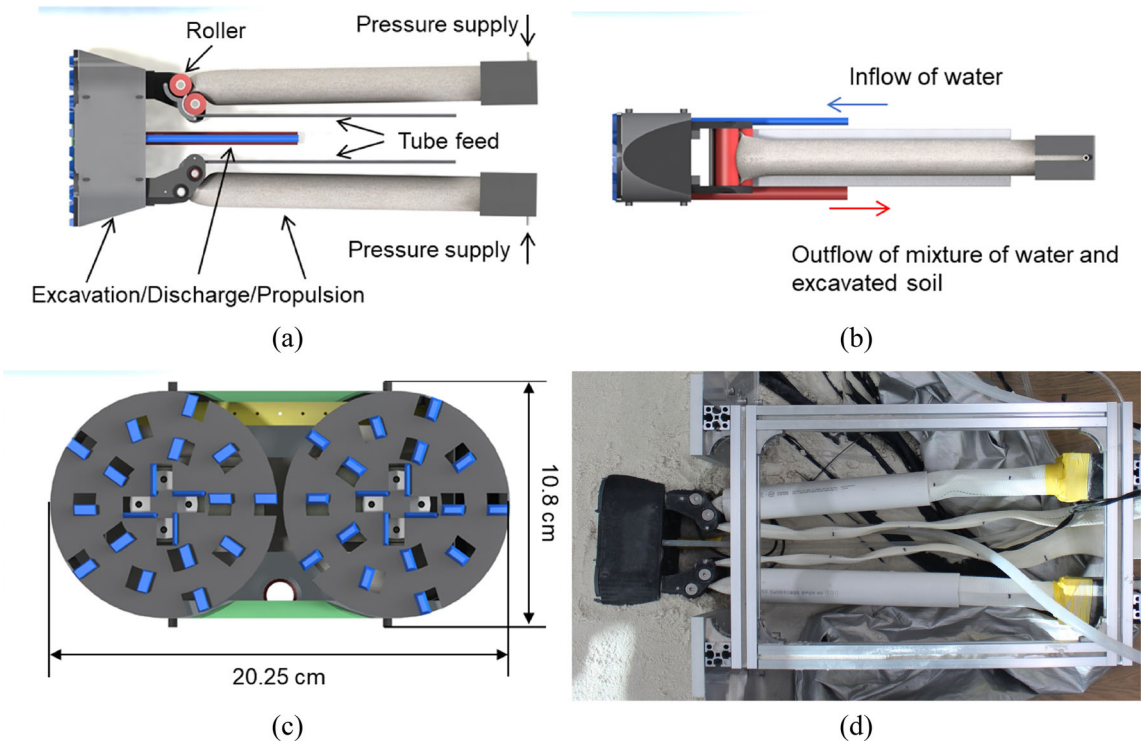


Fig. 3 The prototype of the soft growing excavation robot: **a** top view, **b** side view, **c** front view, and **d** a digital photo of the prototype

prototype RootBot has two cutterheads. Each cutterhead has a diameter of 10 cm, and hence, the excavation module is 20 cm wide and 10 cm high. Detailed descriptions of these three modules follow.

2.2.1 Propulsion (growing) module

The propulsion module has two tube feed systems, in which the tube feeds are individually controlled as well as their air pressure supplies, as shown in Figs. 2 and 3. The propulsion module delivers the propulsion force (or thrust force) to the excavation module through air pressure. And, it manages steering and growth during excavation and advancement by controlling the feeds of the two tubes.

The tube material needs to be soft and flexible for folding and inflation and for curved and complex routes, and it is preferable to be tough, rugged, and resilient to withstand air pressure for sufficient thrust force. Previously, soft-growing robots have been designed to navigate in air and made of vinyl, rubber, or other soft materials [2, 8, 24]. By contrast, RootBot uses a particular type of tubes made of woven polyester filament yarn with inner polyurethane lining layer to withstand a high air pressure up to 0.7 MPa (Model: YFH-S40-40A, Yooksong Co. Ltd., Republic of Korea). This is the same type used as a fire hose.

Compressed air is supplied to each tube from an air cylinder through a pressure regulator (Fig. 3a). The back ends of tubes are connected to the pressure regulators in the base which individually control the pressure and supply of compressed air to tubes. The front ends of tubes near the excavation module are clenched tightly by the rollers and hence it holds the air pressure with minimal air leakage. There are two winches in the base which control the feeds of the uninflated tubes. As the tube is fed and the tube passing the rollers inflates, RootBot advances by growing. Individual controls of air pressure and tube feeds enable full navigation and steering of the excavation module. The thrust force is applied by air pressure in the tubes to the excavation module, such that buckling nor complicated curved path hardly interferes the force delivery.

2.2.2 Excavation module

In the excavation module, two circular cutterheads are rotated by two electronic servo motors (Model: XW540-T260-R, Robotis Inc., Republic of Korea), as shown in Fig. 4. Regulation of DC voltage inputs to the motors allows the controls of the rotational speed of the cutterheads. The diameter of each cutterhead is 10 cm, and the excavation face is 20 cm wide and 10 cm high. The length of the excavation module is 8.7 cm, combining the cutterheads, electronic motors as well as the discharge chamber. When excavating, the two cutterheads rotate in an opposite direction to balance the net rotational moment of the excavation module. When they rotate in the same

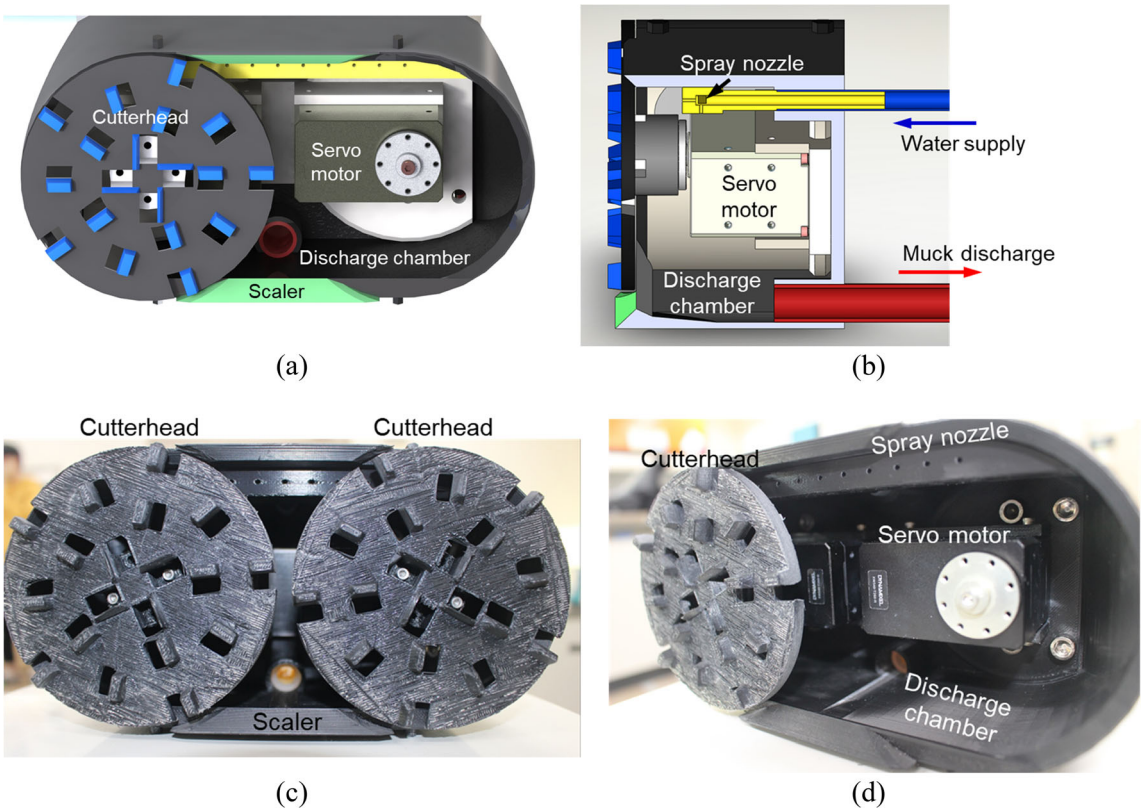


Fig. 4 The excavation module: **a**, **b** 3D drawing of the module, and **c**, **d** digital photos of the fabricated excavation module. Note that the diameter of the cutter head is 10 cm

direction, the torque can be generated, which can cause a rotation of the whole excavation module and the soft tubes.

The cutterhead houses several protruding scrapers and openings for muck discharge, following the designs of the widely used TBM cutterheads. As the cutterhead rotates, the scrapers cut the soil face, and the scraped soils moves to the discharge chamber behind the cutterhead through openings. Additionally, two scalers at the top and bottom of the excavation module scrape the soil in the blind spots uncovered by two circular cutterheads. This design of cutterheads can be adjustable and readily modified according to ground conditions.

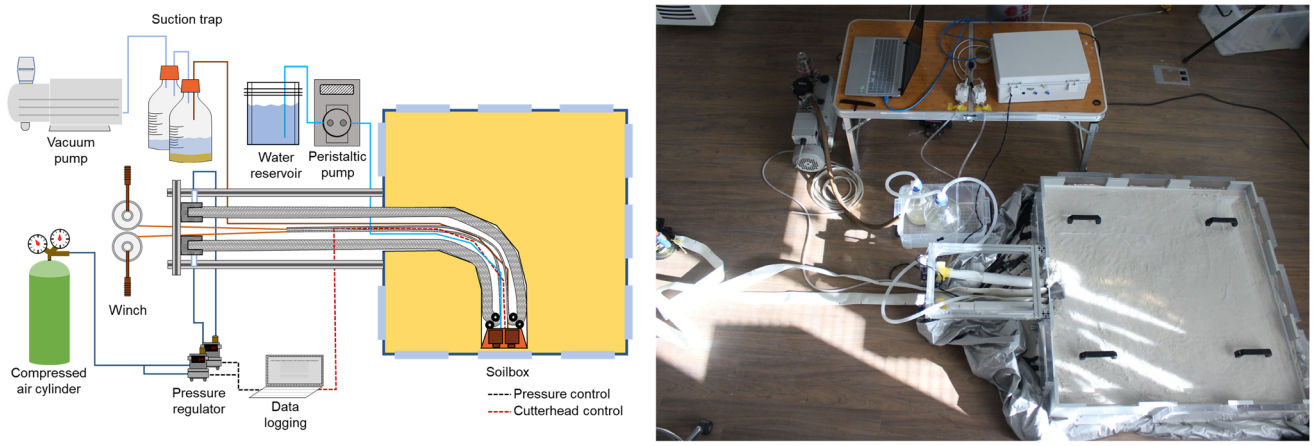
2.2.3 Discharge module

Continuous excavation requires the removal of the scraped soils accumulating in the discharge chamber, which is called muck discharge in tunneling engineering. Generally, common TBM uses a screw conveyor to discharge and convey back the excavated muck from the mixing chamber, and thereby to maintain a continuous excavation speed [18]. The discharge chamber houses two plumbing lines to facilitate the removal of the excavated soils. One plumbing line, located at the top, sprays water through a spraying nozzle, and the other, located at the bottom, retrieves the mixture of excavated soils and water by suction (Fig. 4). Therefore, a water pump and a vacuum pump are connected to those plumbing lines with silicon tubes as peripheral equipment. Thereby, during excavation water is sprayed for lubrication inside the excavation module casing. Simultaneously, a suction trap collects the discharged muck by suction. This discharge system allows continuous excavation without accumulation of excavated soils in the discharge chamber.

3 Experimental study

3.1 Mock-up experiment setup

The mock-up experiments were designed to examine and demonstrate the excavation capability of the developed prototype RootBot. The mock-up experiments used a meter-scale soil box which is 1 m wide, 1 m long, and 0.25 m high. Figure 5 shows the prototype RootBot and the mock-up experiment setup. Moist fine sand was wet-compacted at a water content of 11–14% by hand tamping in the soil box, and the final compacted void ratio and relative density ranged 0.83–0.84 and 56–59%, respectively. Table 1 lists the properties of the soil used in the mock-up



(a) (b)

Fig. 5 Mock-up excavation test setup: **a** the schematic diagram of mock-up excavation experiments, and **b** a digital photo of the test setup

Table 1 Index and basic properties of sand used

Property	Value
Specific gravity G_s	2.65 g/cm^3
Minimum void ratio e_{min}	0.68
Maximum void ratio e_{max}	1.06
Grain size parameters	
D_{15}	0.075 mm
D_{50}	0.124 mm
D_{75}	0.170 mm
Void ratio e	0.83–0.84
Relative density D_r	56–59%
Water content w	11.3–14.4%

experiments. The shear strength of the wet-compacted sand ranged ~ 72–101 kPa, which were measured by the laboratory vane shear test and by the fall cone test. After compacting the moist sand to a target height of 20 cm, an acrylic lid was placed, which applied an overburden stress of approximately 1 kPa. This acrylic lid was transparent, which allowed monitoring of advancement of RootBot.

Peripheral equipment was prepared for the water supply, vacuum suction, compressed air supply, tube feed, cutterhead rotation control, as well as the data acquisition, as shown in Fig. 5a, b. A water pump supplied water to spray on the excavated soils in the discharge chamber. A vacuum pump was connected to the discharge plumbing through a suction trap, which provided suction to retrieve the muck from the discharge chamber. A compressed air was provided from a compressed air cylinder to the growing tubes through the pressure regulators. Herein, in the mock-up experiments the air pressure was regulated with the pressure regulators up to 300 kPa. The computer program controlled the cutterhead rotational speed (revolutions per minute, or RPM) from 10 to 30 RPM during excavation. Two winches regulated the feeds of tubes under pressurized conditions.

During excavation, the rotational speeds of the cutterheads, motor torques and the air pressures in the tubes were measured. The motor torque was estimated based on the supplied power ($P = V{\cdot}I$) during excavation. It is worth noting that the measured torque values should be used for qualitative comparison, not quantitative assessment, as a motor with a high-gear ratio is inherently affected by gearbox friction. In addition, the advancement and route of the RootBot were also monitored from the top by using a digital single-lens reflex (DSLR) camera (100D, Canon Inc., Japan).

3.2 Excavation scenarios

This study demonstrates four excavation scenarios while increasing the turning angle from 0° to 90° for directional excavation, as shown in Fig. 6: (a) straight excavation, (b) directional excavation with a moderate steering angle less than 30°, (c) high-curvature directional excavation with a sharp steering angle more than 60°, and (d) quarter-turn excavation in aid of a brief retraction motion.

First, the RootBot advances along a straight route. This straight excavation was carried out while supplying equal air pressures at the left and right tubes but varying the cutterhead rotational speed and the thrust with air pressure (Fig. 6a). The advanced distance with time was monitored for the cutterhead rotational speed of 10 RPM and 30 RPM and the air pressure of 100 kPa and 120 kPa, respectively.

Second, the RootBot advances along a moderately curved route. This directional excavation with a gentle turn was demonstrated by making a difference in the air

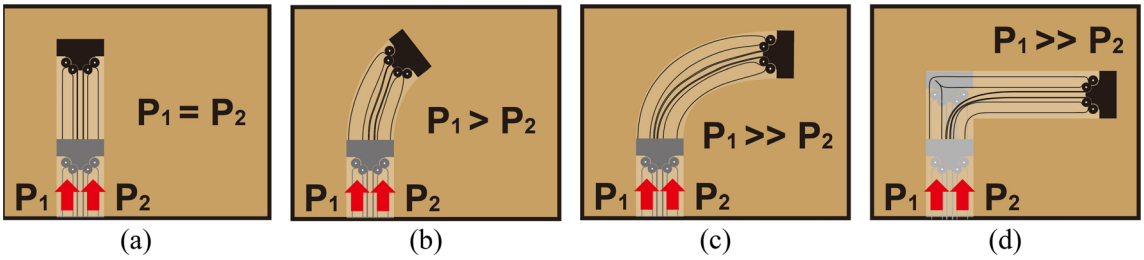


Fig. 6 Excavation scenarios: **a** straight excavation, **b** directional excavation, **c** high-curvature directional excavation, and **d** quarter-turn excavation

pressures at the left and right tubes (Fig. 6b). In the beginning of this test, the straight excavation started with the equal air pressure of 50 kPa. Then, the left-side air pressure increased to 100 kPa while the right-side air pressure was kept at 50 kPa, which steered the direction of the excavation module to the right.

Meanwhile, a pressure reduction at one tube can facilitate steering of the excavation module. Therefore, when combined, a simultaneous action of a pressure increase at one tube and a pressure decrease at the other tube can induce a sharper turn. This action can demonstrate a shaper turn if the tube feeding is manipulated together. In the third scenario, the RootBot shows high-curvature directional excavation with a sharp turn greater than 60°, as shown in Fig. 6c. From the equal air pressure of 50 kPa in the both tubes, the right-side pressure decreased to ~ 10 kPa while the left-side pressure increased to ~ 100 kPa, which was expected to generate a sharp rotation to the right.

The fourth scenario combines the tube feed controls with the pressure controls, and attempts a quarter-turn or a 90° rotation with backward movement (retraction), as shown in Fig. 6d. Expectedly, the winch can manipulate releasing and rewinding of the tube, which allows the advancement and retraction of the excavation module. After straight excavation at the equal air pressure of 100 kPa, the excavation module retracted by rewinding the both tubes. Thereafter, the right-side air pressure gradually decreased to less than 50 kPa while the left-side air pressure increased and then was kept at 300 kPa. This was expected to create a very sharp turn, close to a right-angled turn. Note that the directional excavations (Scenarios 2-to-4) involved pre-trimming of the overlying soil layer along the designed path to aid in visualization.

4 Experiment results

4.1 Scenario 1: Straight excavation test

The excavation performance or efficiency of the developed RootBot can be assessed with the excavation speed (or penetration rate) during the straight excavation tests, which is calculated by dividing the advanced distance with the excavation time. Figure 7 shows the results of the straight excavation test while the equal air pressure was maintained in the both tubes. Owing to the two circular cutterheads, the resulting excavation face is shaped as a rectangle with a width of ~ 20 cm and a height of ~ 10 cm (Fig. 7a). The excavation test results highlight the effects of the rotational speed and the thrust on the advancement speed, and demonstrate that the greater thrust and the higher rotational speed led to the faster advancement speed. Between Run 1 and Run 2, as the thrust increased by increasing the air pressure from 100 to 120 kPa, the advancement speed increased from 0.72 ± 0.15 mm/min to 0.82 ± 0.17 mm/min, by ~ 14%. Between Run 2 and Run 3, as the rotational speed decreased from 30 to 10 RPM under the air pressure of 120 kPa, the advancement speed decreased from 0.82 ± 0.17 mm/min to 0.52 ± 0.08 mm/min, by ~ 58%. Among four runs, Run 4 with 100 kPa air pressure and 10 RPM showed the slowest advancement speed of 0.28 ± 0.2 mm/min. Whereas, Run 2 with 120 kPa air pressure and 30 RPM excavated at the fastest speed of 0.82 ± 0.17 mm/min. When normalizing the advancement speed change with the increment of pressure or cutterhead rotational speed, increasing the thrust force with air pressure appears to be more effective in enhancing the excavation performance than increasing the cutterhead rotational speed. The maximum advancement speed is presumed to differ with the ground condition as well as with the operating controls, such as the air pressure in the tubes and the rotational speed of the cutterhead.

The rotational torque appeared to stay consistent in the range of 1–2 N m, and it was hardly affected by the rotational speed and air pressure for the tested ranges. For each excavation session, the rotational torque was measured to begin as high as 1.5–2 N m and then gradually decreased to 1 N m. This decreasing trend was possibly attributable to the sprayed water in the excavation module smeared into the compacted sand and softening the ground.

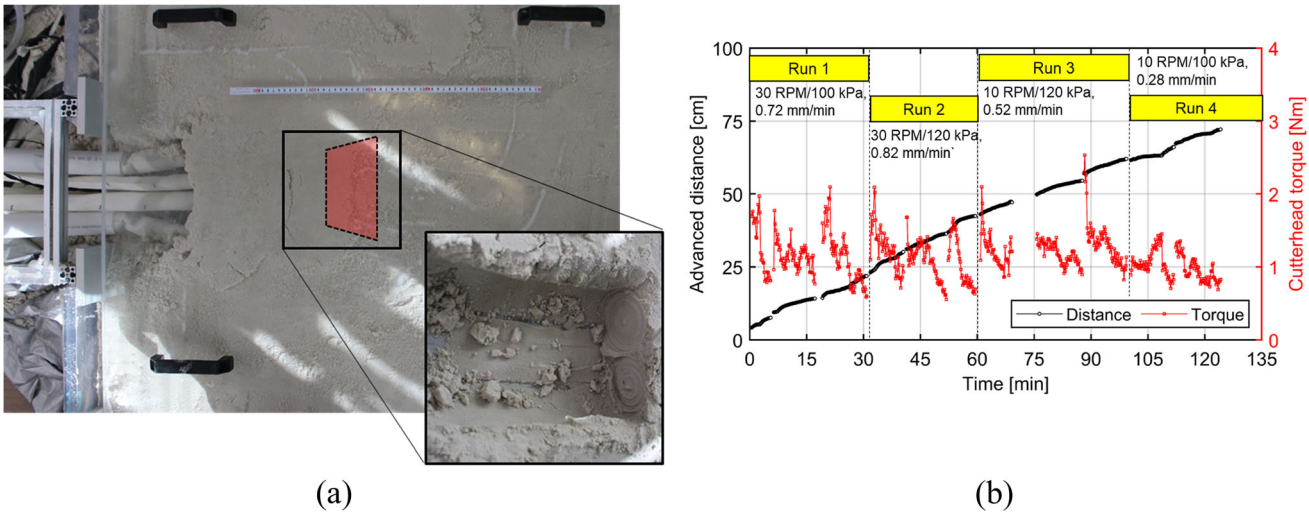


Fig. 7 Result of straight excavation: **a** a photo of the test environment, and **b** the advancement result. The test was carried out with the cutterhead rotational speed of 30 RPM and the air pressure of 100 kPa

4.2 Scenario 2: Directional excavation test with a moderate steering angle

The directional excavation was carried out while the cutterhead rotation speed was kept constant at 30 RPM, and the results are shown in Fig. 8. In Stage 1 (from Point A to B in Fig. 8a), the left- and right-side pressures were initially kept the same at ~ 45 kPa for straight excavation about 10 cm. Thereafter, during excavation without stopping, the left-side pressure increased to 90 kPa and the right-side pressure was elevated to 60 kPa. As a result, the RootBot moderately rotated with the steering angle of 18° (Point C in Fig. 8a). Additional pressure adjustment at Point D resulted in an additional 23° turn. Upon the rotation of the excavation face with a final steering angle of 41°, the both side pressures were equalized at 60 kPa, which led to a straight excavation (from Point D to Point E in Fig. 8a). The resulting route is shown in Fig. 8b, and the cutterhead torque ranged ~ 0.3–0.6 N·m during this directional excavation. This cutterhead torque was lower than that in the straight excavation, attributable to the pre-trimming of the overlying soil layer in this demonstration.

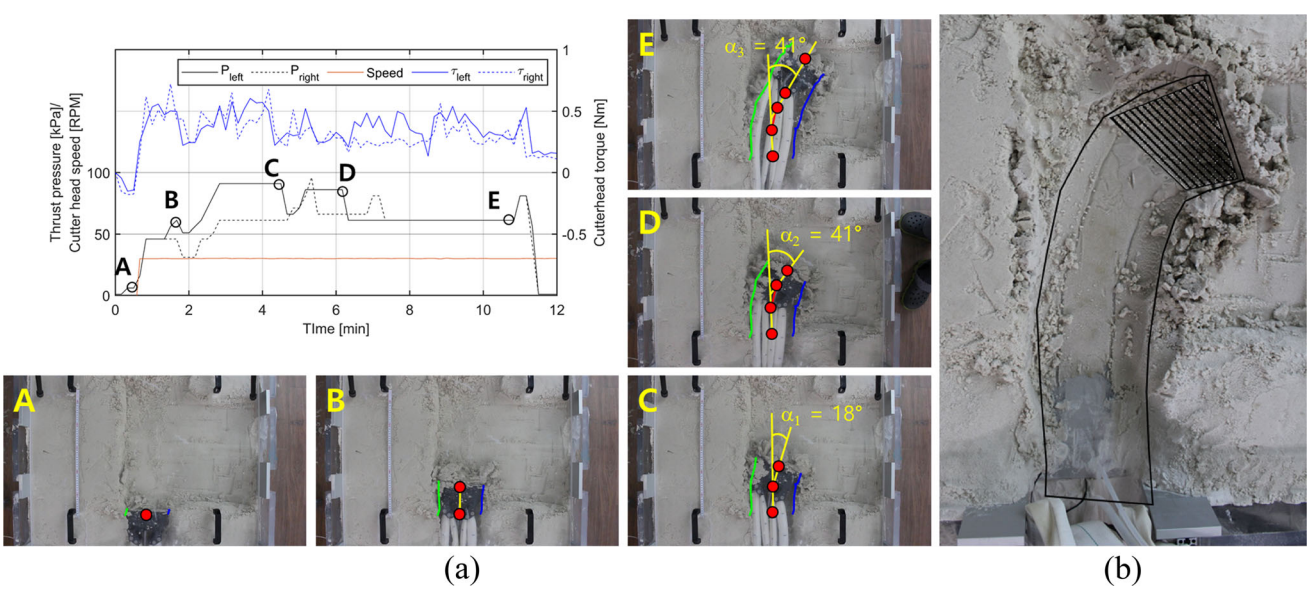


Fig. 8 Result of directional excavation: **a** the pressure manipulation, cutterhead rotational speed, and cutterhead torque with digital photos taken during excavation, and **b** resulting excavation path

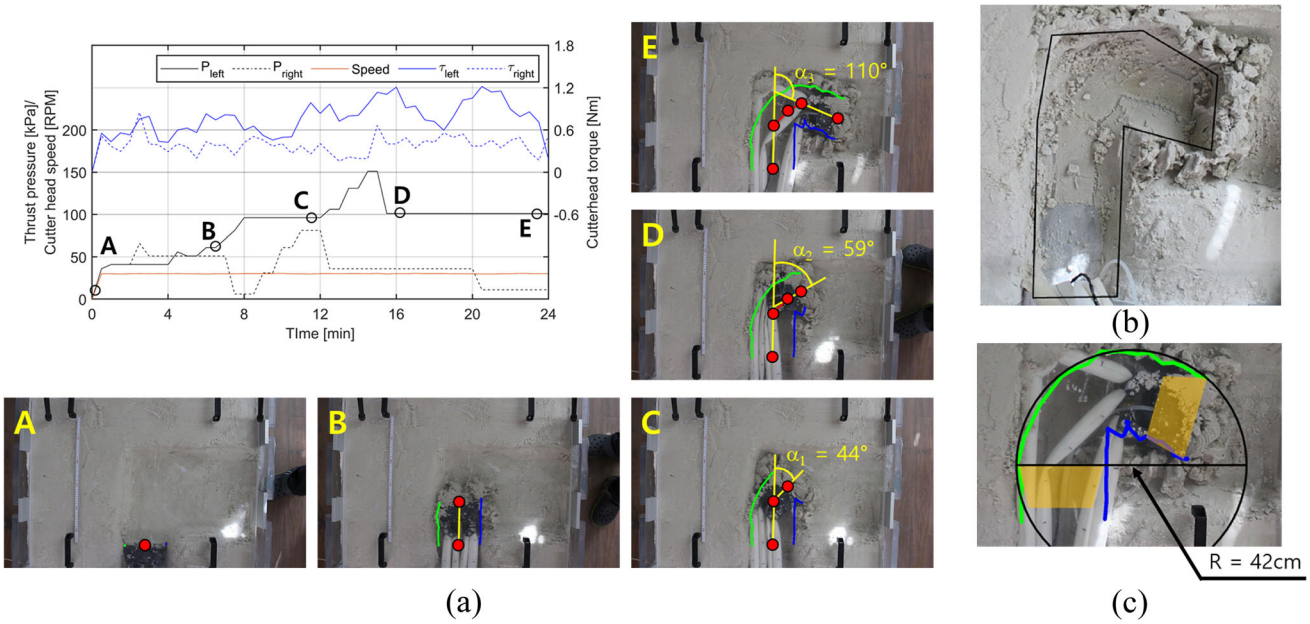


Fig. 9 Result of high-curvature directional excavation: **a** the pressure manipulation, cutterhead rotational speed, and cutterhead torque with digital photos taken during excavation, **b** resulting excavation path, and **c** excavation radius

4.3 Scenario 3: High-curvature directional excavation with a sharp steering angle

A sharper steering angle was attempted with the prototype RootBot, and the results are shown in Fig. 9. A straight excavation was performed with the equally balanced pressure at ∼ 50 kPa (from A to B in Fig. 9a). Immediately after Point B, a sharp steering was induced by simultaneously increasing the left-side pressure to ∼ 100 kPa and decreasing the right-side pressure to ∼ 10 kPa. As a result, the excavation face steered right by 44° (Point C in Fig. 9a). Thereafter, keeping the pressure difference greater than 50 kPa during advancement caused further rotation, and hence the steering angle gradually increased to greater than 90° (from C through D and to E in Fig. 9a). Due to this high curvature in the route, the left-side tube became buckled (Point E in Fig. 9a). Nevertheless, the buckling hardly affected the excavation and advancement of the RootBot as it grew from the feeds of soft tubes. This result demonstrates that the RootBot can grow and navigate through complicated and curvy routes while excavating soils. Figure 9b shows the route or path after excavation, which confirms a directional excavation with a steering angle sharper than 90°. The curvature radius of the curved excavation was estimated to be 42 cm, which was 2.1 times greater than the excavation module width.

4.4 Scenario 4: Quarter-turn excavation in aid of a retraction movement

The last demonstration was conducted while taking advantage of the RootBot's ability to retract, as shown in Fig. 10. First, the straight excavation was performed by RootBot with equally balanced air pressure at 100 kPa (From A to B in Fig. 10a). Then, while the air pressure was balanced and then reduced to ∼ 80 kPa, the excavation module was retracted for approximately 20 cm by pulling back both the left- and right-side tubes with the winches (From B to C). This retraction action pulled back the cutterheads from the excavated soil face, and hence the torques at the cutterheads were reduced close to zero. Herein, we chose the retracted distance to be greater than the half of the excavation width at minimum because this would facilitate a quarter turn (90° rotation) of the excavation module in place without overcutting around soil walls. Following Scenario 3, a sharp rotation of the excavation module was induced by simultaneously increasing the left-side pressure to ∼ 300 kPa and decreasing the right-side pressure to less than ∼ 50 kPa. Subsequently, excavation resumed at a steering angle of 56° (Point D). The ensuing advancement with the pressure difference induced further rotation to 109° (Point E in Fig. 10a). Note that the pressure in the right-side tube from Point C to Point E was intermittently raised up to adjust the posture of the excavation module. The final steering angle was approximately 110°, similar to that of Scenario 3 (Fig. 10b). The curvature

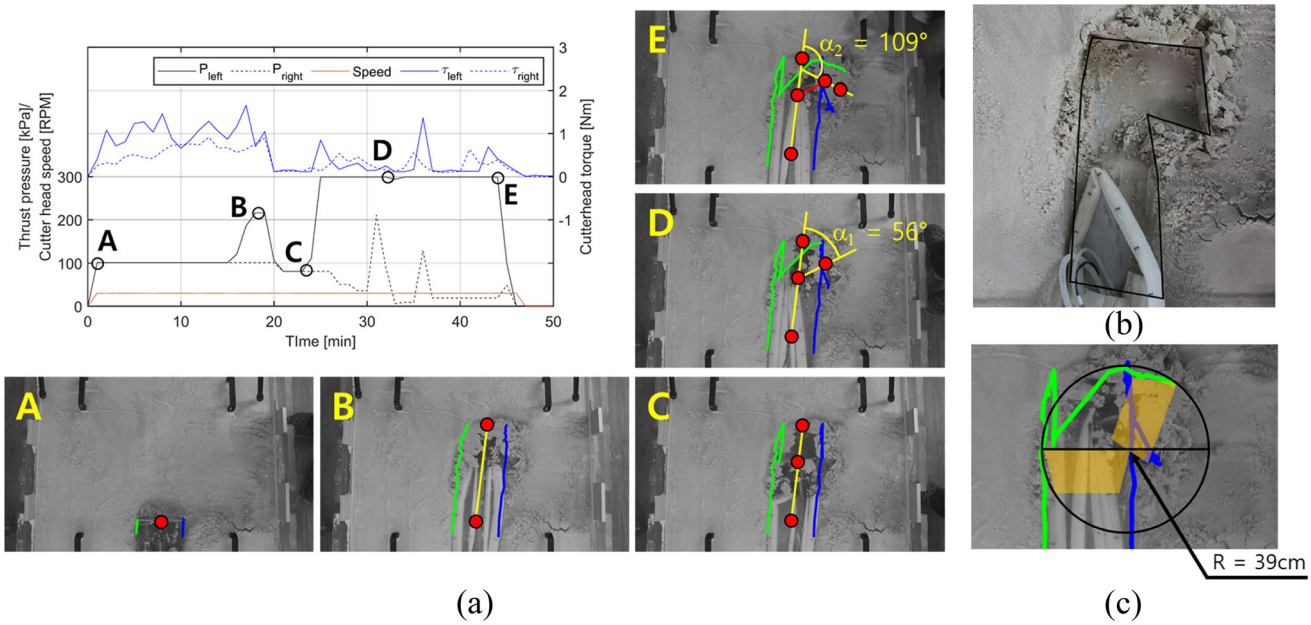


Fig. 10 Result of high-curvature directional excavation with retraction: **a** the pressure manipulation, cutterhead rotational speed, and cutterhead torque with digital photos taken during excavation, **b** resulting excavation path, and **c** excavation radius

radius of the curved excavation was estimated to be 39 cm and 1.95 times greater than the excavation module width (Fig. 10c). This result shows that the curvature radius with the retraction was smaller than that without the retraction.

5 Discussion

5.1 Comparison of steering and directional excavation performance

The curvature ratio is widely used to quantify the path curvature that a robot or a device can steer relative to its size while moving along that curved path. The curvature ratio, defined as the ratio of the minimum curvature radius possible to the width or diameter of the equipment, can be used as an indicator to the steering capacity of an excavation equipment. Accordingly, this section compares the curvature ratio to compare the directional excavation performance of RootBot with the other directional excavation robots as well as the commercial TBMs, as shown in Table 2 [7, 24, 25, 27]. For an instance, the commercial TBMs which were used in urban construction practices are reported to have a curvature ratio greater than 9 [7]. Whereas, the soft-growing robots with a small diameter and a steering capability appears to have the lower curvature ratios less than 5, which highlights their high-curvature directional excavation. The RootBot achieved a curvature ratio of 1.95 in a wet-compacted sand. Similarly, one soft robot showed the curvature ratio of 2 in a granular medium [27], which does not excavate but penetrates into a medium without discharging mucks. The other soft robots had the values of 7.6 in a dry sand [24] and 10.44 in a saturated sand [27].

Table 2 Comparison of steering angle for RootBot and other robotic excavation systems

Equipment and site (References)	Module width or diameter (cm)	Curvature radius (cm)	Curvature ratio (–)[a]
RootBot (this study)	20	39	1.95
Fast, steerable, burrowing soft robot [24]	6	19 (46)[b]	3.17 (7.6)[b]
Self-growing soft robot inspired by plant roots [25]	5	10	2
SEAVO [27]	16	167	10.44
TBM S75, Klong Daan Cable Tunnel [7]	427	4800	11.24
TBM S48, Klong Phra Khanong Cable Tunnel [7]	320	3000	9.38

[a]The curvature ratio is defined as the ratio of the curvature radius to the width or diameter of the excavation equipment

[b]The minimum value measured in air is shown. The minimum value measured in sand is indicated in the parentheses

5.2 Implications

The RootBot is a new robotic system that incorporates a cutterhead-driven excavation design of TBMs and a locomotive part inspired by the growth mechanism of plant roots. The proposed prototype of the RootBot has shown the following distinctive features. (a) The RootBot's growth is facilitated by material feeds to the front tips and is not limited by the excavation routes. (b) This soft growing excavation robot is capable of a directional excavation at a sharp turning angle with a curvature radius of less than two times the excavation head width. (c) It can retract and move backward by rewinding the tubes, which allows pull-back of the excavation module and a quarter-turn excavation at any point along a straight path.

The scalability is another benefit of the RootBot as it can be readily extended for a specific task. The maximum allowable air pressure is 0.7 MPa in the presented prototype, but if necessary, different tube materials can be used for a higher operating air pressure, and even different fluids such as water or oil can be used to apply thrust force. The prototype's excavation section is rectangular, and the different shape and size of the excavation module can be easily fabricated and examined. Also, in the discharge system, more viscous fluid with a dispersant may be beneficial to transport excavated mucks and minimize particle clogging in transport tubes and pipes in the discharge system.

The RootBot is expected to play a role in practices associated with ground excavation and soil removal in inaccessible and confined spaces, primarily owing to the capability to excavate at a high curvature and navigate complicated paths through steering by pressure difference and retraction movement. As an example, RootBot can navigate through Y-joint or T-joint pipe lines as well as straight pipe lines and perform pipe repair works. If necessary, it can excavate and remove dirt and soils deposited and plugged in sewer pipes and navigate to the arrival point for required jobs. The RootBot can also steer and excavate in ground while avoiding pre-existing underground structures and obstacles. For an example, drilling is very challenging when unexpected boulders and core stones are encountered. Whereas, the RootBot can simply retract and find a curved route to pass by those unexpected obstacles.

However, further research and development is warranted to address the limitations and improve the system's performance. First, the locomotion of the RootBot is limited to a planar motion because the prototype moves with two soft growing tubes. For an instance, it is not allowed to move up and down when it advances horizontally. However, the current version of RootBot can be readily extended to three-dimensional excavation through various ways, which includes but not limited to cutterhead rotation adjustment, addition of one or more vines and cutterheads, and implementation of non-rotating excavation methods such as waterjet nozzles. And, horizontal excavation at a deeper depth and vertical excavation should be further examined for field application. Second, the excavated space can collapse as there is no casing or lining installation. In the rear part, grouting or casing with a rigid structure may be necessary as in tunnel construction. Third, this study only demonstrates the excavations in relatively soft sandy soils with the rectangle-shaped excavation module. A variety of prototyping and experiments are required with different excavation section, cutterhead design, cutterhead size, and various ground types. Forth, the execution of multiple steep turns may be challenging due to the limitation of soft growing tubes in buckling control. Therefore, a feedback system, including motion sensors, can be effective to improve post-buckling maneuverability and enhance its ability to reach the predefined target.

Furthermore, the effect of controlling variables, such as pressure, cutterhead rotational speed, and torque, on excavation performance need to be quantitatively assessed. Such quantitative assessment can be properly carried out for straight excavation, but it is quite challenging for curved excavation. The curved excavation typically takes much longer time than the straight excavation, and it is also heavily affected by the operational strategy. The presented results, though it is qualitative, still provide valuable insights into the potential of high-curvature directional excavation using the proposed robotic system. Further research is required to establish robust and quantitative relationships among variables for assessment and optimization of excavation performance.

6 Conclusion

This study presents a new soft-growing robotic system, RootBot, which exploits inspiration from plant root growth to excavate with a steering capability in complex and confined spaces. This RootBot is mainly composed of three parts: the propulsion (growth) module, which manages the locomotion of the robot, the excavation module with two circular cutterheads with servo motors, and the discharge module with a water spray tube and a suction tube, which removes the scraped soils and hence enables continuous excavation. A prototype of Roobot was fabricated with a 20 cm wide and 10 cm high rectangular excavation module. A series of mock-up experiments examined the excavation and steering capability of the prototyped RootBot under various excavation scenarios. The test results show that the prototype RootBot effectively excavates and advances at a speed of 0.82 mm/min in wet-compacted

sand, and it demonstrates its steering capability at high curvature with a curvature ratio of less than 2. Further research is warranted to demonstrate its versatile capability in environments with various excavation sections, cutterhead designs, cutterhead sizes, and ground types. With proper further development, the RootBot would be a valuable tool in geotechnical and urban engineering practices, including directional excavation of grounds, soil removal in inaccessible and confined spaces, and navigation through a complicated, curved route to pass by unexpected underground obstacles.

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Author contributions All authors contributed to the study's concept and design. GH and DS were responsible for robot design and fabrication, experimentation, data collection, and analysis. The first draft of the manuscript was written by GH, with inputs and feedbacks from all authors. JR and TK were responsible for reviewing and editing the manuscript, as well as providing research guidance. All authors have reviewed and approved the final manuscript.

Data availability All the data used in this study are available upon reasonable request to the corresponding author. Additionally, the data will be soon published online in shared data repositories.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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